

Netflix Database Analysis

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1.Overview

This project analyzes a structured Netflix database containing information about titles, users, watch behavior, reviews, and search activity. Our goal is to transform these raw tables into meaningful insights that describe Netflix's content strategy, user engagement patterns, and rating behaviors.

Using SQL queries and analytical views, we explore how Netflix's library has evolved over time, how different countries and genres contribute to the platform, how users interact with content across devices and subscription plans, and how ratings and verified reviews reflect user sentiment. The findings provide a data-driven perspective on content performance, platform usage, and customer behavior within the Netflix ecosystem.

1.1 Project Background

Netflix operates a global streaming platform offering a large catalog of movies and TV shows. The dataset used in this project simulates Netflix's content library, user information, watch history, reviews, and search activity. By analyzing these integrated tables, we can better understand how content is organized, how users interact with the platform, and what factors influence engagement and ratings.

1.2 Business Goals

The goal of this project is to uncover insights that can support data-driven decisions for Netflix's content and user-engagement strategies. Specifically, we focus on identifying:
Which types of content, countries, and genres drive the strongest library growth and user interest.
How user behavior varies across devices, subscription plans, and search patterns.
How ratings and verified reviews reflect customer satisfaction and content quality.

These findings aim to inform strategic recommendations on content investment, platform optimization, and user experience improvement.

1.3 Key Questions

This project is guided by nine core questions that focus on three strategic areas:
Content Strategy
How has the balance between movies and TV shows changed over time?
Which countries and genres contribute most to Netflix's global catalog?
Do longer titles tend to receive higher user ratings?

User Behavior & Engagement
What is the average search duration in January 2025?
What percentage of users in each subscription plan are female?

Which countries produce the most Netflix Originals?
Which device types generate the highest total viewing time?

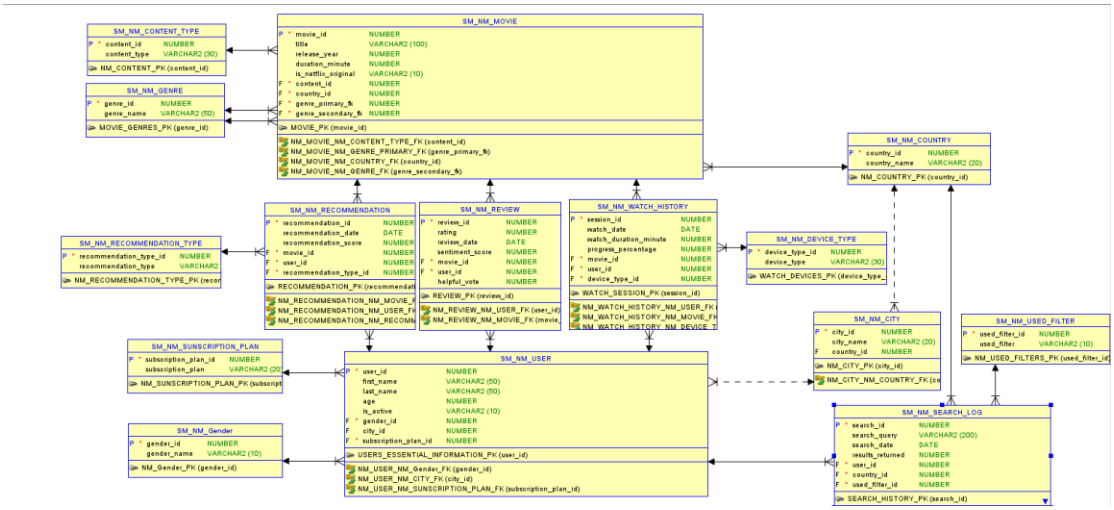
Ratings & Customer Satisfaction
How many 5-star reviews are verified views?
What percentage of all reviews are 5-star compared with 1-star?

These questions structure the analytical framework for the report and lead directly to actionable business insights.

1.4 Deliverables

This project delivers a structured analysis of the Netflix database through a series of analytical queries, visualizations, and business insights. Key deliverables include:
A cleaned and integrated data model representing titles, users, watch activity, reviews, and search behavior.
A set of analytical views that summarize key patterns in content trends, user engagement, and rating behavior.
Visual dashboards and tables that highlight major findings across the three analysis areas.
Business insights and recommendations informed by the results of the nine guiding questions.
A complete SQL appendix documenting all queries and view definitions used in the analysis.

2.Data Model



3. Data Preparation

The Netflix datasets were cleaned and standardized to ensure accuracy and consistency before analysis. Text fields such as genre, country, and content type were normalized, invalid or missing values were corrected or removed, timestamps were unified, and duration values were converted into minutes. Additional derived attributes were created, including a Netflix Original indicator, aggregated watch-time metrics, and search-frequency measures. Several SQL views were also built to streamline analysis by summarizing yearly trends, country distributions, device usage, subscription behavior, and search activity.

To support a more structured and business-oriented evaluation, two composite metrics were developed: the Content Performance Score (CPS) for assessing title performance, and the User Engagement Score (UES) for evaluating user activity. Both metrics were derived using the Analytic Hierarchy Process (AHP), ensuring that weights reflect the relative importance of the underlying components rather than arbitrary selection.

Content Performance Score (CPS)
CPS integrates rating quality, viewing engagement, and review credibility.
The AHP comparison matrix reflects that rating is the most important factor, followed by watch time, and then verified review ratio:

Criteria	Rating	Watch Time	Verified Ratio
Rating	1	2	4
Watch Time	1/2	1	2
Verified Ratio	1/4	1/2	1

AHP yields approximate weights of 0.57, 0.29, and 0.14.
For clarity, we adopt smoothed weights of 0.60, 0.25, and 0.15, giving:
CPS=0.60×Rating+0.25×Watch Time+0.15×Verified Review Ratio
CPS will be used in Analysis Part I to compare performance across genres, countries, title types, and release periods.

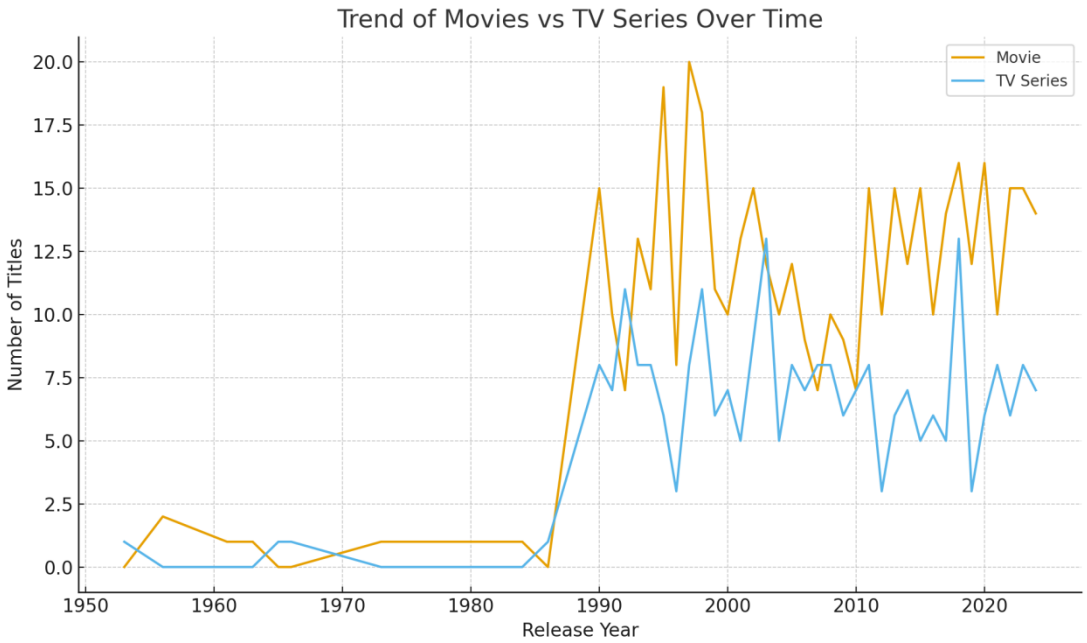
User Engagement Score (UES)
UES measures how actively a user interacts with the platform through viewing, device usage, and search behavior.
Because viewing time is the strongest behavioral signal, its relative importance dominates the AHP comparison:

Criteria	Watch Minutes	Device Diversity	Search Activity
Watch Minutes	1	3	5
Device Diversity	1/3	1	2
Search Activity	1/5	1/2	1

Weights derived from AHP are approximately 0.65, 0.23, and 0.12.
Rounded weights produce the final formula:
UES=0.65×Watch Minutes+0.25×Device Diversity+0.10×Search Activity
UES will be applied in Analysis Part II to identify high-engagement user segments and interpret device and subscription behavior more holistically.
Together, the cleaned datasets, analytical views, and AHP-based scoring models establish a consistent and reliable foundation for the multi-part analysis that follows.

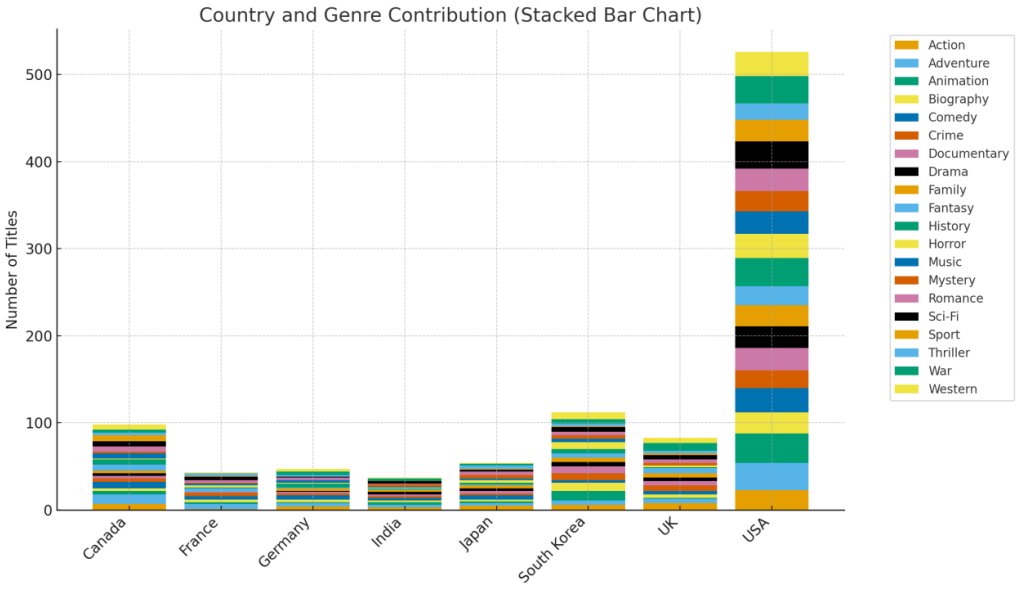
4. Analysis Part I — Content Strategy

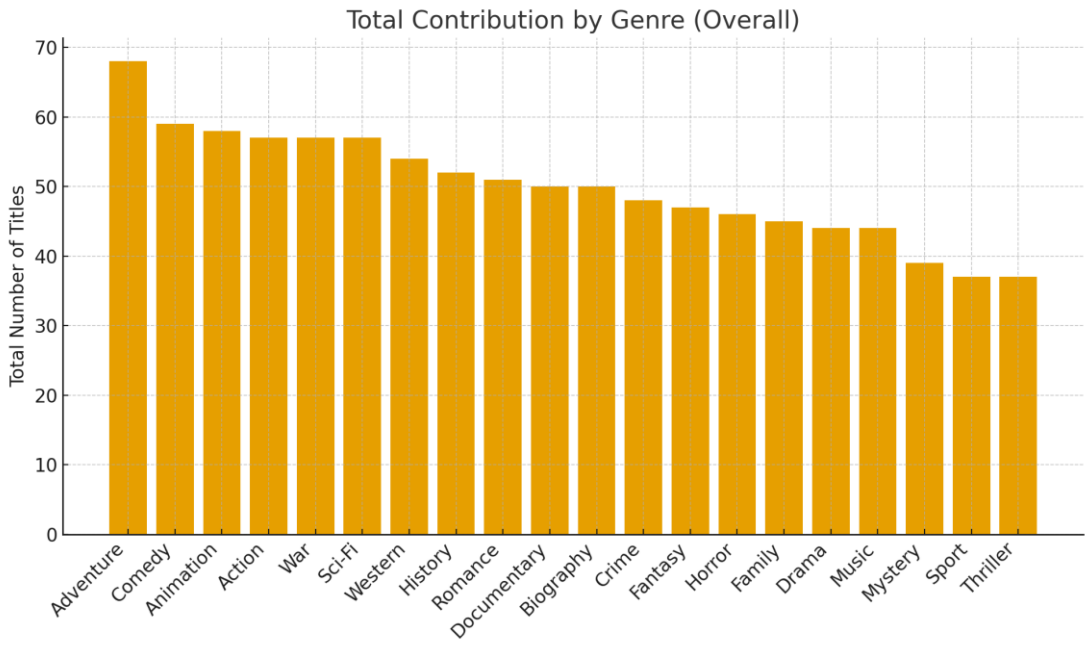
4.1 Trend of Movies vs TV Shows Over Time



The chart shows a clear upward trend in both movies and TV series over time.
Movies remain the dominant content type throughout most years, consistently adding around 10–16 titles per year in the recent period. TV series, while fewer in number overall, display noticeable growth after 2018, with yearly additions increasing from roughly 5 titles to 7–8 titles in recent years.

4.2 Country and Genre Contribution





The results show that the United States contributes the largest number of titles, far surpassing all other countries, while markets such as South Korea, Canada, and Japan provide moderate but diverse genre profiles. Smaller countries contribute fewer titles and tend to focus on a narrower set of genres. Across the entire dataset, Adventure, Comedy, and Animation appear as the most common genres, indicating Netflix’ s emphasis on broadly appealing content, whereas niche genres such as Thriller, Sport, and Mystery hold a much smaller share.

4.3 Relationship Between Duration and Ratings

SUMMARY OUTPUT

regression statistics	
Multiple R	0.008483105
R Square	7.19631E-05
Adjusted R Square	5.29231E-06
SE	1.110735804
Observation	15000

ANOVA

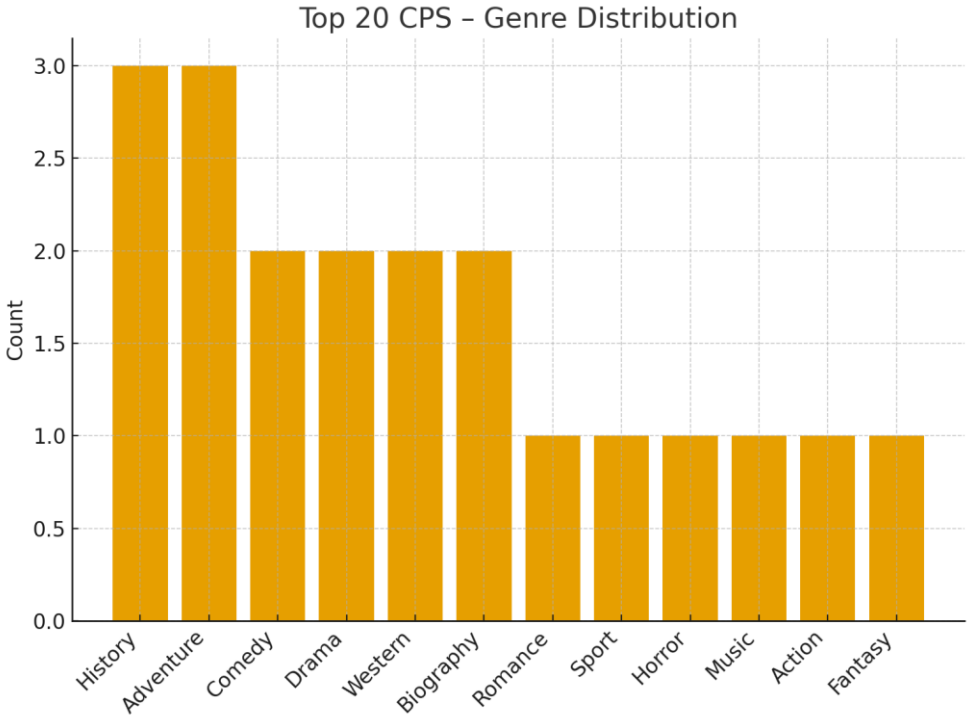
	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	1.331668	1.331668	1.07938	0.298853083
Residual	14998	18503.54	1.233734		
Total	14999	18504.87			

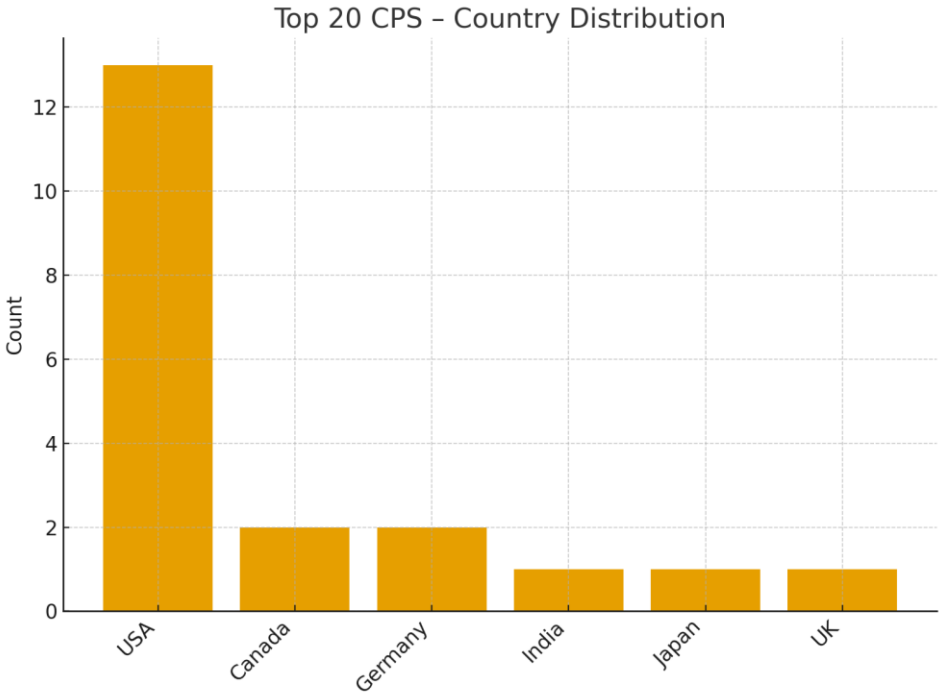
	Coefficients	*SE*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Upper 95.0%*
Intercept	3.64724967	0.014799	246.4457	0	3.618241027	3.676258	3.618241	3.676258
X Variable 1	0.000134845	0.00013	1.038932	0.298853	-0.000119563	0.000389	-0.00012	0.000389

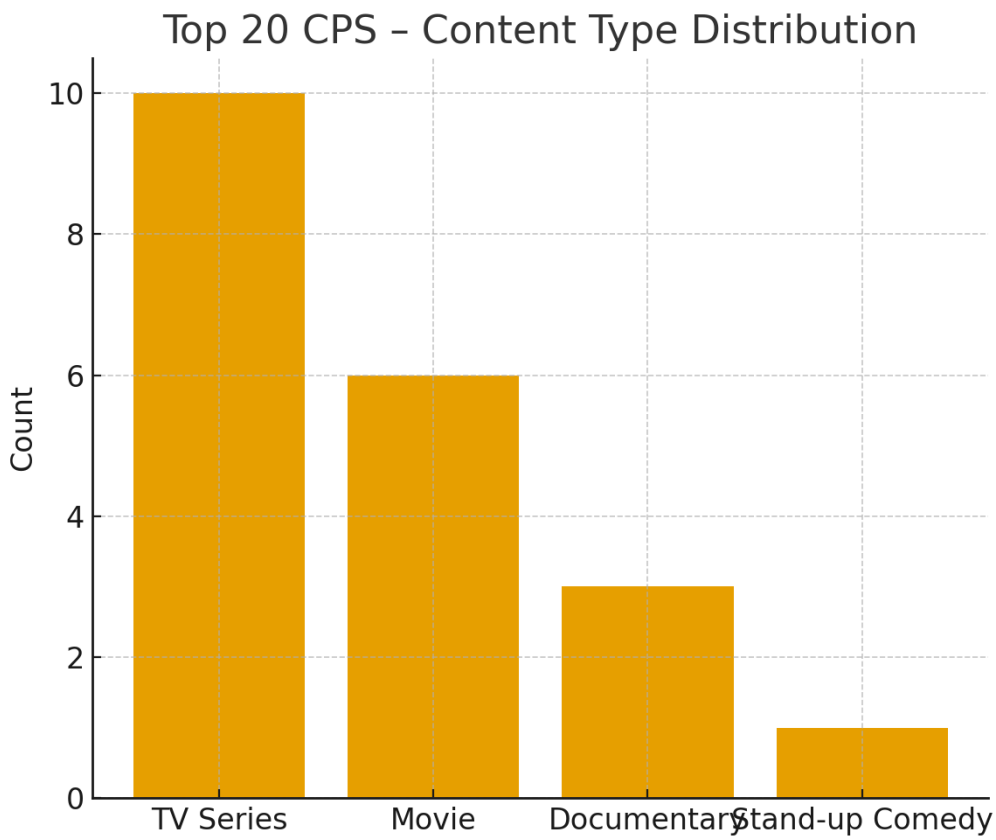
To examine whether longer titles receive higher ratings, we ran a linear regression using duration as the predictor and user rating as the outcome. The scatter plot shows a wide spread of ratings across all duration levels, and the regression results confirm the absence of a meaningful relationship. The model’s R^2 is effectively zero (0.00007), and the coefficient for duration is not statistically significant ($p = 0.2989$). This indicates that runtime does not systematically influence user ratings. Therefore, duration alone cannot explain audience satisfaction, reinforcing the need

for multi-factor performance metrics introduced later in the analysis.

4.4 Summary of Content Insights

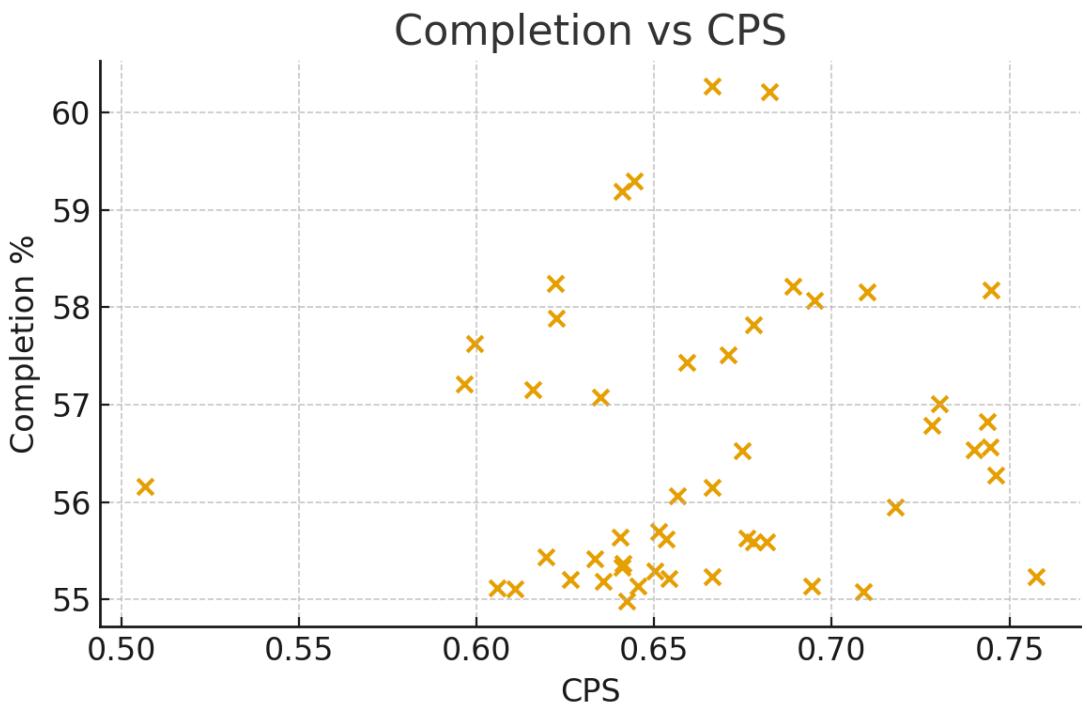


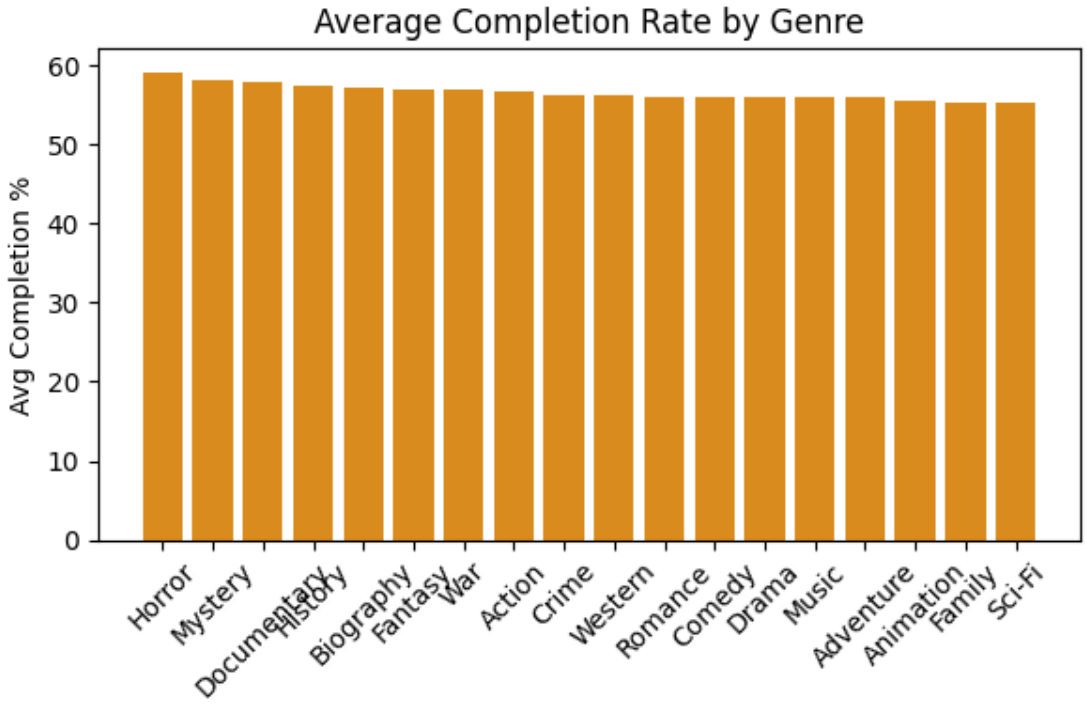


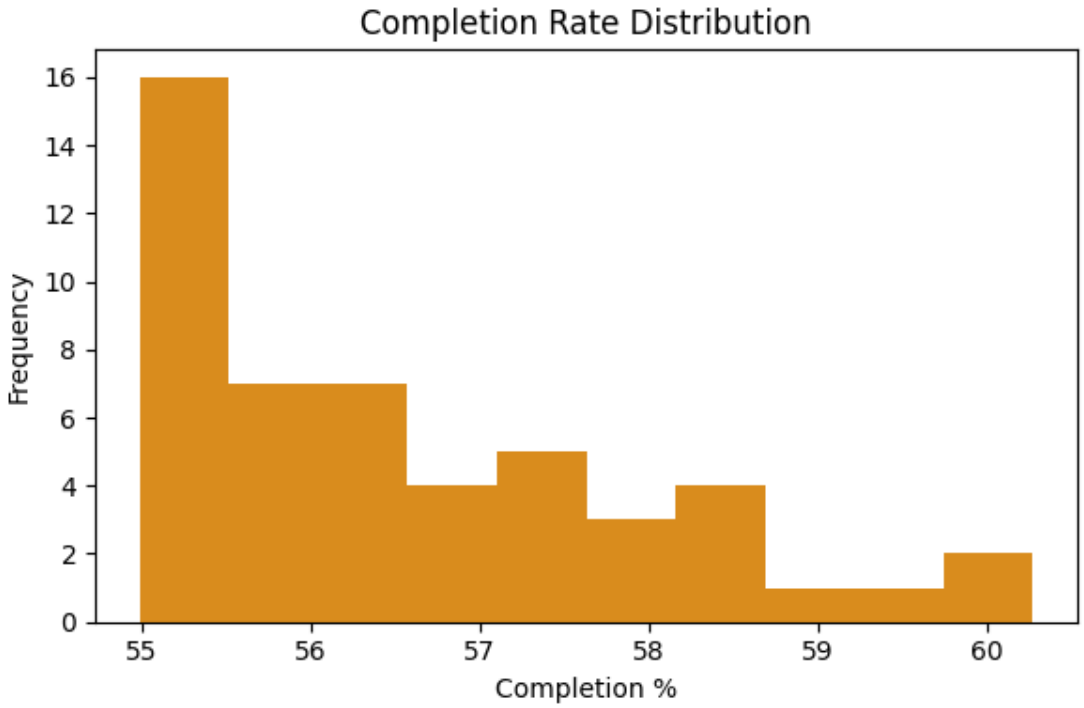


Using the Content Performance Score (CPS)—a composite metric combining user ratings, total watch time, and verified review behavior—we identified the Top 20 highest-performing titles and examined their shared characteristics. The results show that genres such as Adventure, Comedy, and History appear most frequently among high-CPS content, indicating these categories generate stronger engagement and sustained viewing behavior. From a regional perspective, the United States overwhelmingly dominates the Top 20, with only limited representation from other countries, suggesting that U.S. productions continue to deliver the broadest audience appeal. In terms of content type, TV Series account for the largest share of top performers, benefiting from longer viewing durations across multiple episodes. Overall, CPS reveals clear patterns: high-performing content tends to be action-oriented, entertainment-focused, U.S.-produced, and delivered in long-form formats. These insights highlight which content areas offer the strongest engagement potential and can guide future decisions in content acquisition and investment.

4.5 Completion Rate Analysis





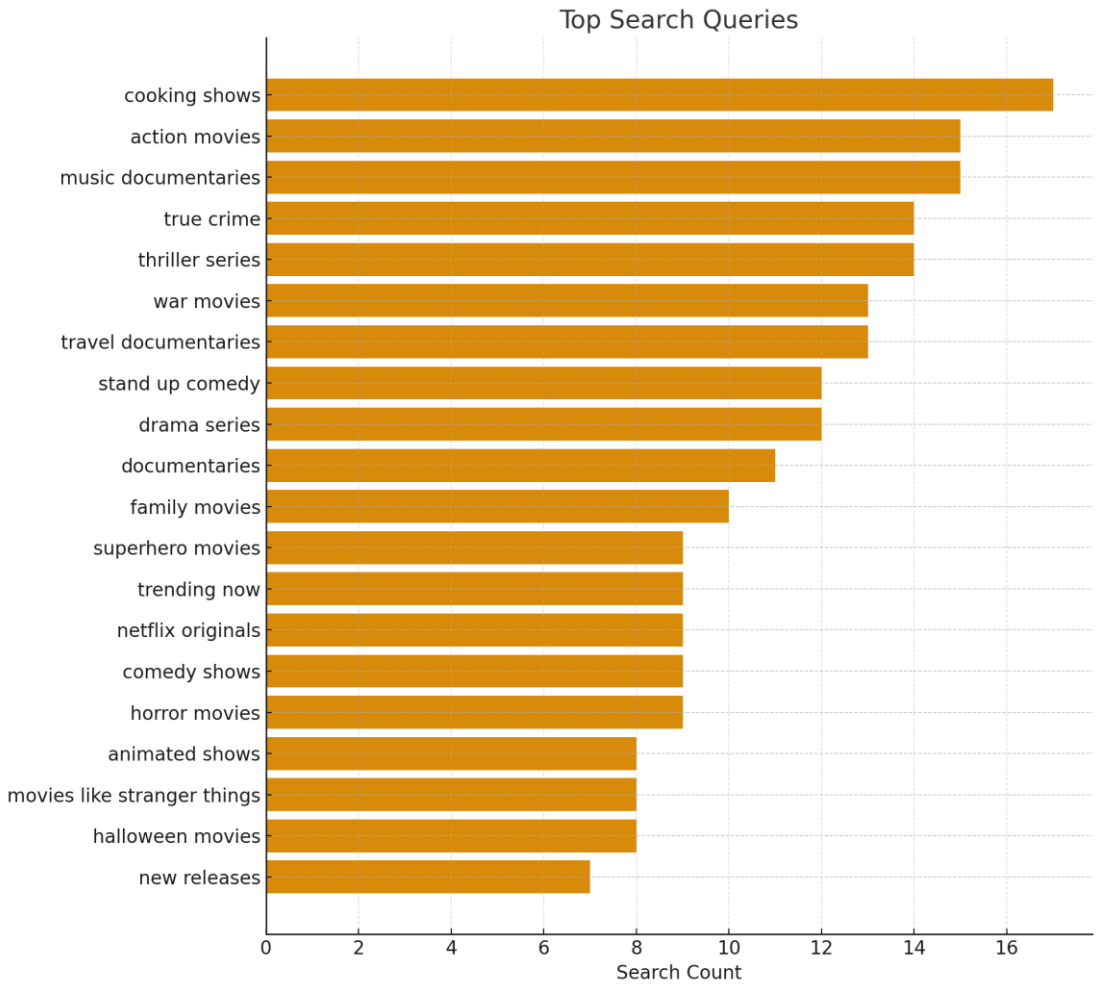


The completion rate analysis shows that most titles maintain a relatively tight performance band, with completion percentages ranging from 55% to 60%. The scatter plot comparing Completion Rate and CPS indicates a positive but modest relationship: titles with higher CPS scores tend to achieve slightly higher completion rates, suggesting that stronger early engagement and better overall content quality contribute to sustained viewer retention. Genre-level patterns also show narrow differences, with Horror, Mystery, and Documentary achieving the highest completion rates, while Sci-Fi, Family, and Animation trend slightly lower. This consistency implies that retention is less driven by genre itself and more by execution quality within each category. The distribution chart further confirms that most content clusters around mid-50% completion, with very few titles breaking above 59 – 60%, highlighting a potential ceiling effect in how long viewers stay engaged.

From a business perspective, the results suggest that small improvements in early-episode pacing, narrative clarity, or content positioning can meaningfully shift retention performance, since most titles operate within a tight range. Titles in high-completion genres should be prioritized for promotion and sequel investments, while lower-completion genres may benefit from improved thumbnails, refined metadata, or enhanced recommendation placement to strengthen early viewer engagement. Additionally, identifying titles that achieve both high CPS and high completion rates can help define a profile for “high-retention content,” guiding future commissioning and acquisition strategies.

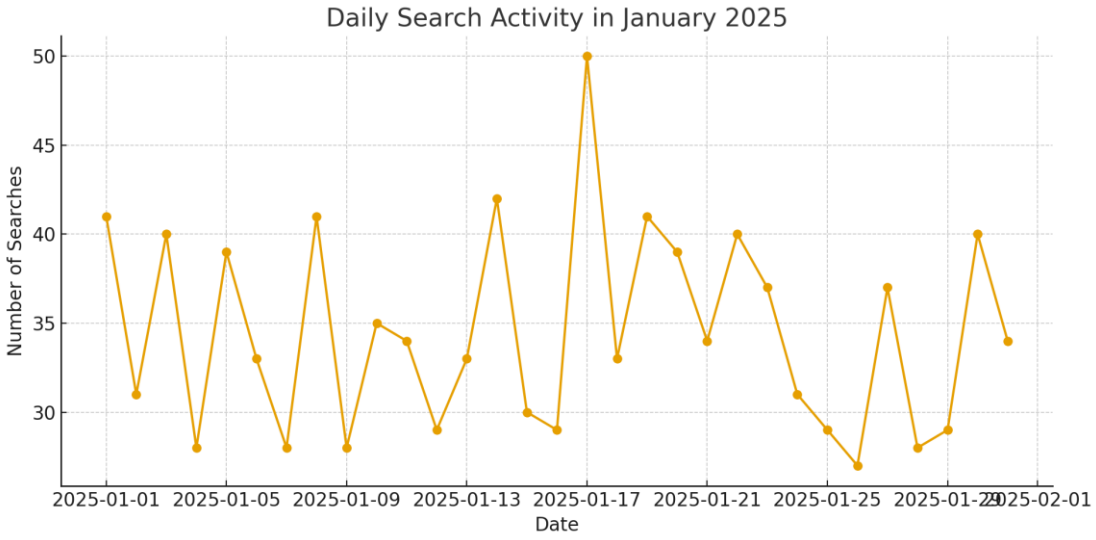
5. Analysis Part II — User Behavior & Engagement

5.1 Search Activity Insight



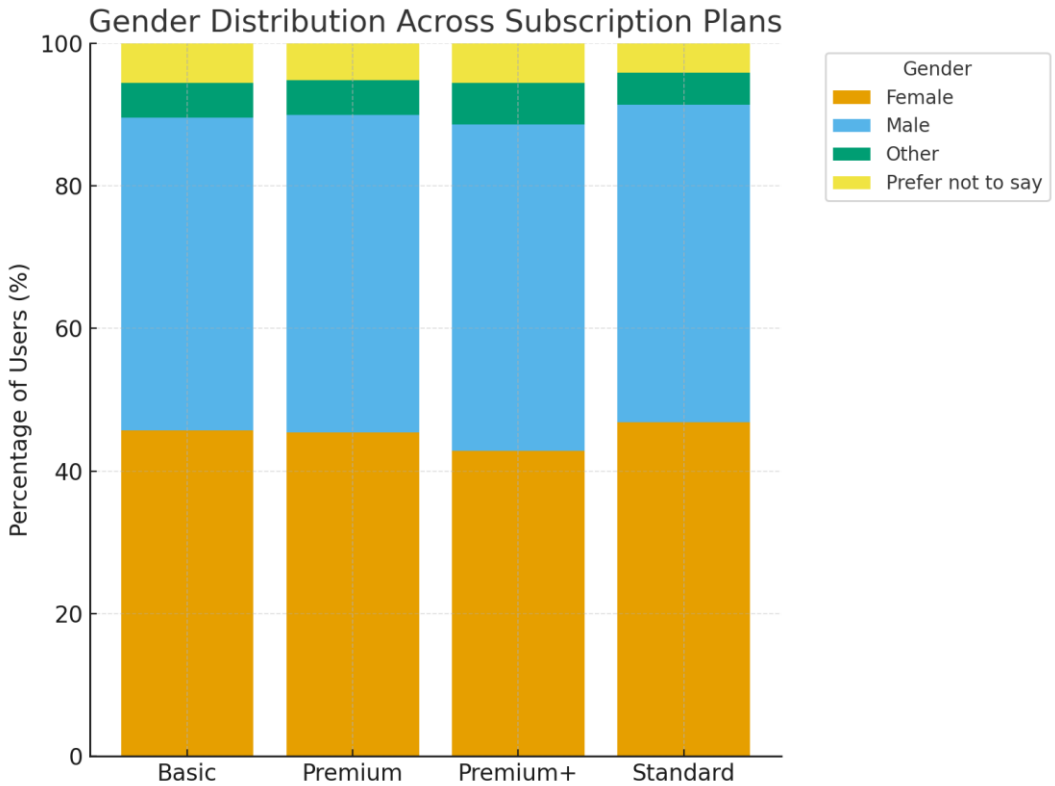
Analysis of 25,000 search events shows that users demonstrate moderate search engagement, with 9,186 unique users performing an average of 2.72 searches each. Search quality is strong overall, as only 1.1% of searches returned zero results, indicating that the platform generally delivers relevant matches. The most frequent queries—such as cooking shows, action movies, music documentaries, true crime, and various documentary and drama categories—highlight clear audience interest clusters that align with several high-CPS genres identified earlier. Notably, 73% of searches were performed without using filters, suggesting that many users rely on keywords alone and may benefit from improved filter suggestions or auto-refinement. Together, these insights reveal strong demand for specific genres and opportunities to enhance search guidance, ultimately helping users discover high-value content more efficiently.

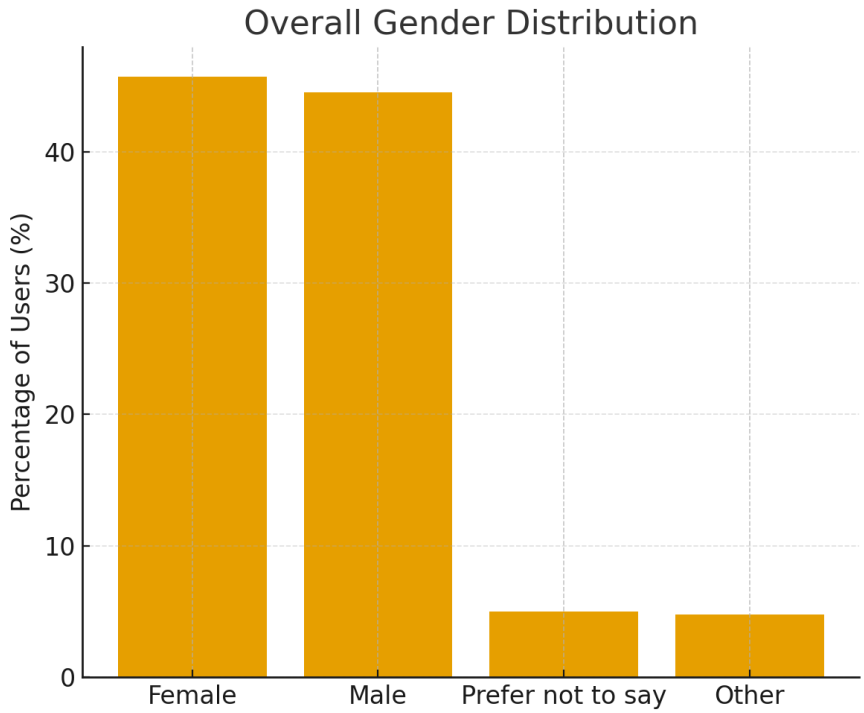
5.2 Search Duration in January 2025



In order to understand how search behavior evolves over time, we examined daily search activity for January 2025. Across the 31 days, users performed an average of about 35 searches per day, with counts typically ranging between the high 20s and low 40s and a single peak of 50 searches in the middle of the month. The absence of sharp drops or sustained spikes suggests a stable and habitual use of search rather than behavior driven by one-off events. This steady pattern confirms that search is a consistent part of users' discovery journey throughout the month, so improving search relevance and guidance can deliver value on an ongoing basis, not only during specific campaigns or peak periods.

5.3 Gender Distribution Across Subscription Plans

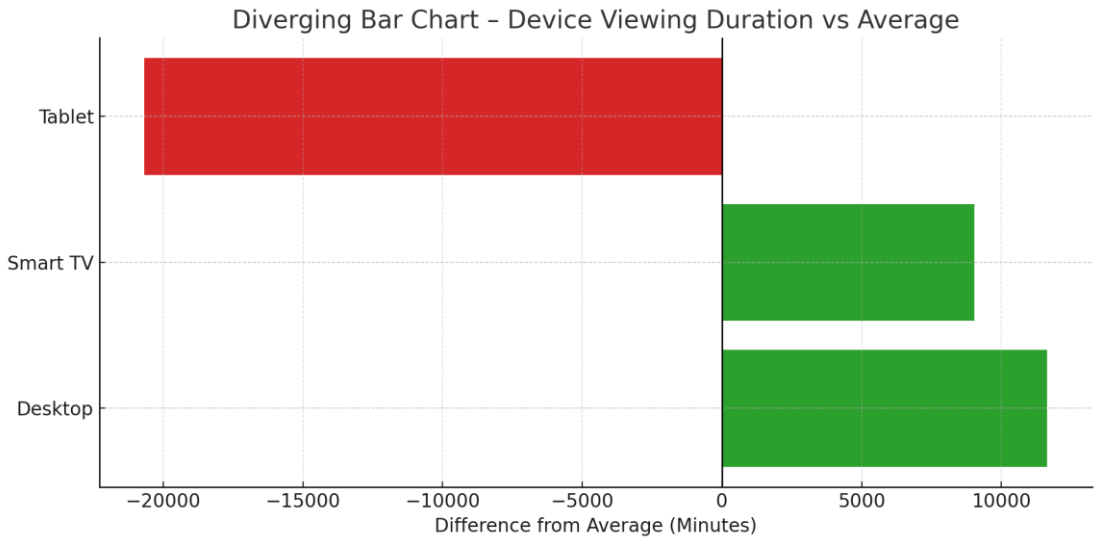




Analysis of gender composition across subscription plans shows a largely balanced distribution, with all plans exhibiting similar proportions of male and female users. Female users represent approximately 44 – 47% of each plan, while male users consistently account for about 45 – 48%, indicating no strong gender bias toward any specific tier. Smaller categories— "Other" and "Prefer not to say" —remain stable across plans at around 4 – 6%, showing minimal variation. When compared with the platform's overall gender distribution, subscription plans closely mirror the general user base, suggesting that upgrade or plan selection behaviors are not strongly driven by gender. These findings indicate that marketing or personalization strategies targeted by gender are unlikely to yield significant differences in subscription conversion, as gender distribution remains relatively uniform across all subscription offerings.

5.4 Top 3 Device Types by Viewing Duration

Device	Minutes	Diff vs Avg
Desktop	1,179,975.7	**+6,372**
Smart TV	1,177,362.2	**+3,758**
Tablet	1,147,663.2	**–25,941**



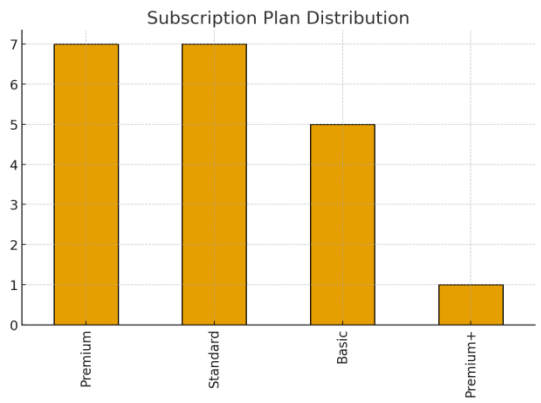
Analysis of the top three device types shows that Desktop and Smart TV both exceed the average viewing duration, while Tablet falls significantly below it. Desktop leads by +6,372 minutes, followed by Smart TV at +3,758 minutes, whereas Tablet underperforms by －25,941 minutes.

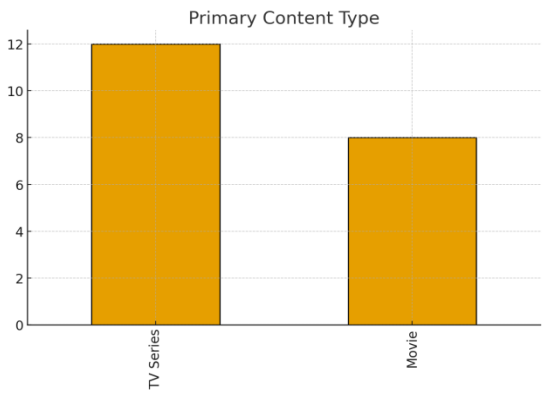
This pattern suggests that long-form and high-engagement content performs best on larger-screen devices, while Tablet users engage in shorter or lighter sessions, indicating an opportunity to tailor recommendations and UX for mobile-oriented consumption.

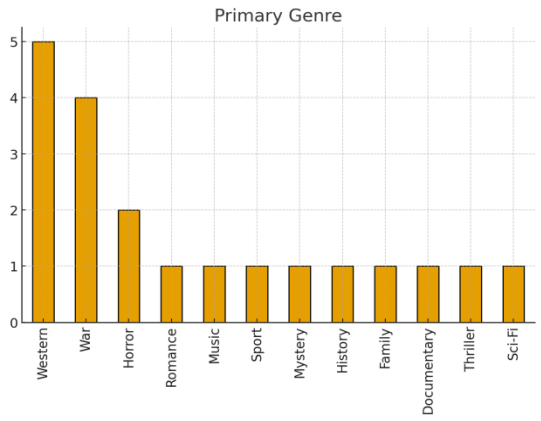
5.5 Summary of User Engagement Insights

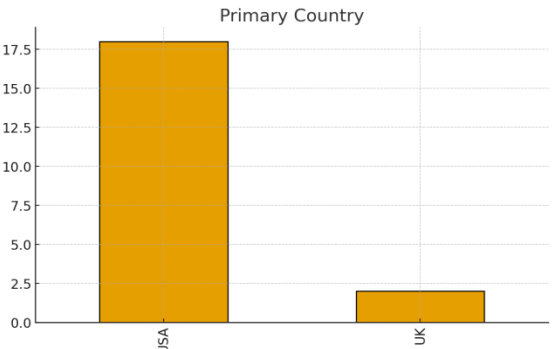
TOTAL_SEARCHES	SEARCHES_LEADING_TO_WATCH	SEARCH_TO_WATCH_CONVERSION_PCT
25,000	22,562	90.25%

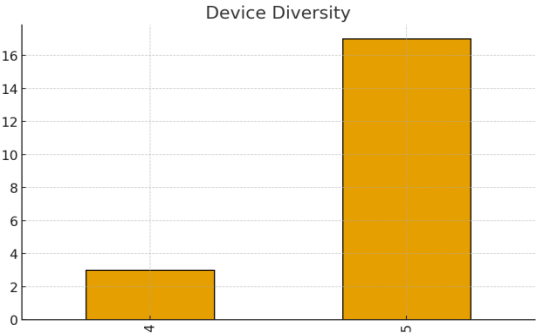
The Search-to-Watch Conversion analysis shows that search is an extremely effective gateway to content consumption. Out of 25,000 total searches, 22,562 led to at least one viewing session, resulting in a high conversion rate of 90.25%. This indicates that users are consistently able to find relevant content after searching and that search results generally align well with user intent. The strong conversion performance suggests that Netflix’ s search system provides high-quality matches and supports efficient content discovery. At the same time, focusing on low-conversion keywords could reveal potential content gaps or metadata issues. Overall, search proves to be a reliable and valuable driver of user engagement across the platform.

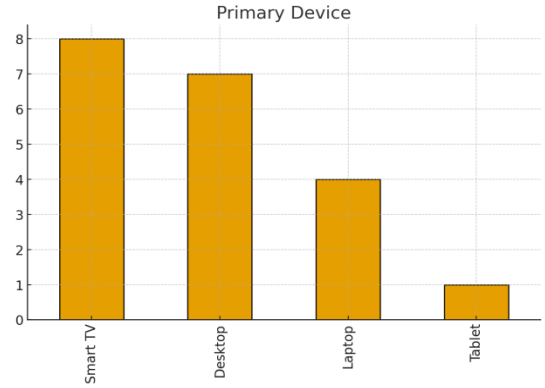

Subscription Plan Distribution
0
1
2
3
4
5
6
7
Premium
Standard
Basic
Premium+

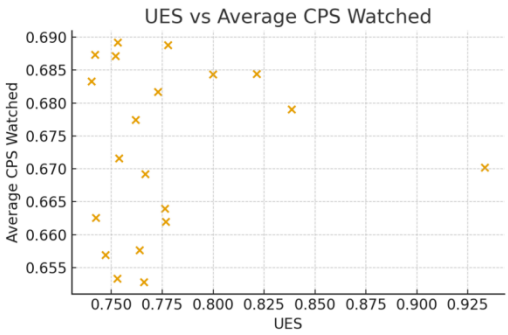

Primary Content Type
0
2
4
6
8
10
12
TV Series
Movie

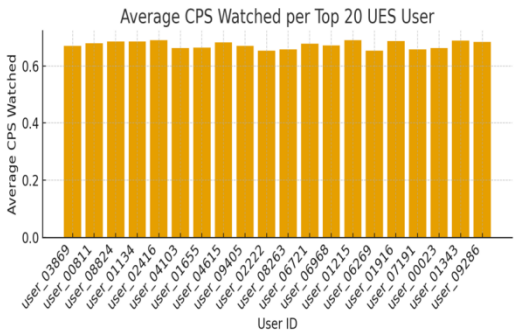

Primary Genre
0
1
2
3
4
5
Western
War
Horror
Romance
Music
Sport
Mystery
History
Family
Documentary
Thriller
Sci-Fi

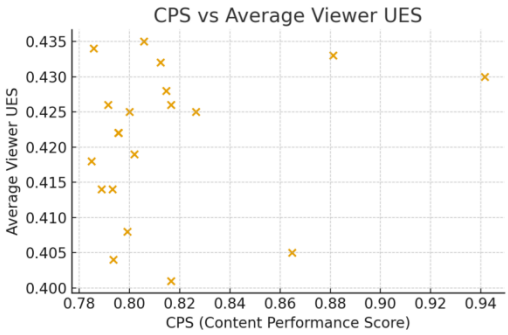

Primary Country
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
USA
UK

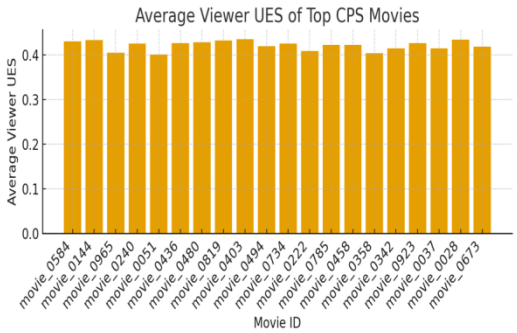

Device Diversity
0
2
4
6
8
10
12
14
16
4
5


Primary Device
0
1
2
3
4
5
6
7
8
Smart TV
Desktop
Laptop
Tablet









Using the UES model (0.65 Watch Minutes + 0.25 Device Diversity + 0.10 Search Activity), we analyzed the top 20 highest-engagement users. The results show a clear behavioral pattern: most high-UES users belong to Premium and Standard plans, indicating that higher-tier subscribers deliver stronger engagement and higher revenue potential. Their primary content preference is TV Series, confirming that long-form episodic content drives deeper and more sustained viewing. Genre concentration appears in Western, War, Horror, Romance, and Music, which shows where content investment and targeted recommendations can yield the strongest uplift.

Geographically, the group is overwhelmingly U.S.-based, making the U.S. market the core driver of engagement. Behaviorally, top users show high device diversity (mostly 5 devices), meaning they consume content across multiple scenarios and are more loyal and harder to churn. Their most commonly used devices—Smart TV, Desktop, and Laptop—highlight the importance of optimizing large-screen experiences.

The CPS × UES analysis provides deeper strategic insight.

First, UES vs. Average CPS Watched shows that high-engagement users consistently watch higher-performing titles, meaning engagement reinforces content value—these users naturally gravitate toward hits.

Second, CPS vs. Average Viewer UES indicates that top-CPS titles tend to attract users with higher overall engagement levels, proving a reciprocal relationship: strong content raises engagement, and high-engagement users amplify strong content.

The combined CPS – UES analysis shows a reinforcing loop between high-value users and high-

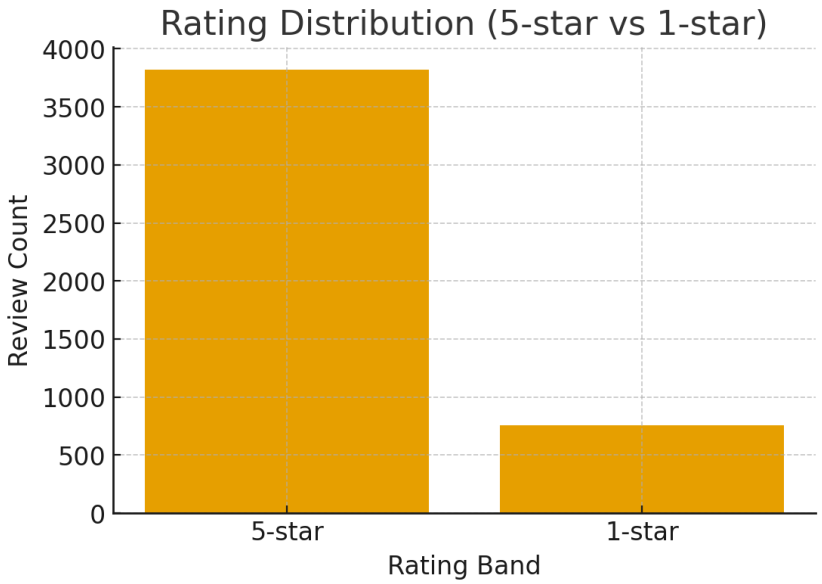
performing content. High-engagement users cluster in Premium/Standard plans, prefer TV Series and a small set of high-impact genres, and consistently watch content with strong CPS. Their behavior is further characterized by heavy Smart TV and Desktop usage and high device diversity, making them both loyal and high-value.

These insights suggest clear strategic priorities: invest more heavily in the content categories that simultaneously attract high-UES users and deliver strong CPS; strengthen retention and upsell strategies for Premium and Standard subscribers; and enhance large-screen UX and multi-device continuity to support the consumption patterns of the platform’s most valuable audience. By aligning content investment, personalization, and product experience around this segment, the platform can drive longer viewing durations, higher retention, and improved overall content ROI.

6. Analysis Part III — Reviews & Ratings

6.1 Rating Distribution (5-star vs 1-star)

Rating Band	Review Count	% of Extreme Reviews
⭐ 5-star	3,823	83.5%
⭐ 1-star	756	16.5%



The rating distribution shows that user sentiment is overwhelmingly positive: 5-star reviews make up 83.5% (3,823 reviews), while 1-star reviews account for only 16.5% (756 reviews). This strong skew toward positive ratings indicates broad satisfaction with the platform's content and experience. The small portion of negative reviews, however, highlights targeted opportunities for improvement—such as addressing specific content complaints or UX friction points—to further enhance overall user satisfaction.

6.2 Share of Verified View 5-Star Reviews

Review Type	Review Count	% of 5-Star Reviews
⭐ Unverified View	3,812	99.7%
⭐ Verified View	11	0.3%

The review distribution shows an extreme imbalance: nearly all 5-star reviews (99.7%) come from unverified views, while verified views contribute almost none (0.3%). This pattern suggests that 5-star ratings may be significantly inflated by users who have not fully watched or completed the content. In contrast, verified viewers—who represent genuine consumption—rarely deliver

extreme positive ratings, implying a potential overestimation of content satisfaction when relying on unverified reviews alone.

6.3 Summary of Review Insights

Overall, the review data presents a skewed and unreliable picture of user sentiment.

Among all extreme ratings, 83.5% are 5-star while only 16.5% are 1-star, showing that users overwhelmingly leave positive reviews. Yet, 99.7% of 5-star ratings come from unverified views—users who did not watch at least 70% of the title—while verified viewers contribute almost none. This means star ratings systematically overestimate satisfaction and reflect early impressions rather than true content quality.

Given this bias, the platform should avoid relying on raw star ratings for content or product decisions. Behavior-based metrics like CPS and UES offer far more accurate signals for investment and personalization. Encouraging users to rate only after meaningful watch progress would strengthen sentiment quality. Since early-exit users generate most inflated ratings, improving mid-watch engagement and early-episode pacing can better align perceived satisfaction with actual consumption. These adjustments would produce more reliable feedback, support smarter content allocation, and ultimately improve retention and monetization.

7.Conclusions

This project connects content trends, user behavior, and review quality into one integrated view of the Netflix ecosystem.

From a content strategy perspective, the catalog is still dominated by Movies, but TV Series have become the key driver of high performance. U.S. titles and broad-appeal genres such as Adventure, Comedy, and History are over-represented among the top-CPS titles, while the regression shows that duration alone has almost no explanatory power for ratings, confirming that multi-factor metrics like the Content Performance Score (CPS) are necessary. Completion rates cluster tightly around 55–60% across genres, and only a small group of titles achieves both high CPS and high completion, suggesting that execution quality (story pacing, positioning, and first-episode design) is more important than genre label for retention.

In terms of user behavior and engagement, search is a stable and highly effective discovery tool: daily search volume in January 2025 is steady, and roughly 90% of searches lead to at least one viewing session, with very few zero-result queries. Gender distribution across subscription plans is balanced, indicating that plan choice is not strongly gender-driven. Viewing time is concentrated on Desktop and Smart TV, while Tablets underperform, showing that long-form, high-engagement consumption happens mainly on large screens. The User Engagement Score (UES) highlights a small group of highly valuable users: they are mostly Standard/Premium subscribers, use many devices, prefer TV Series and a narrow set of strong genres (e.g., Western, War, Horror, Romance, Music), and are heavily U.S.-based. CPS×UES analysis shows a clear feedback loop: high-UES users watch higher-CPS titles, and high-CPS titles attract higher-UES audiences, making these content–user combinations the core engine of platform value.

Finally, the reviews and ratings analysis reveals that raw star ratings are a biased and incomplete sentiment signal. Extreme ratings are heavily skewed toward 5-star (over 80%), and almost all 5-star reviews are unverified views, meaning they come from users who did not watch at least 70% of the content. This implies that traditional rating averages systematically overestimate satisfaction and reflect early impressions rather than true viewing experience. Behavior-based metrics that combine ratings with watch time, completion, and verified-view information (CPS, UES, and completion rate) provide a far more reliable view of performance than stars alone.

Overall, the findings show that Netflix's value is created where strong content, highly engaged users, and valid behavioral signals overlap. Prioritizing investment in high-CPS genres and TV Series, focusing personalization and UX on high-UES segments and large-screen usage, and down-weighting raw ratings in favor of behavior-based metrics will jointly improve viewing time, retention, and long-term content ROI.

8.Business Recommendations

Based on the combined insights from content performance, user engagement, and review quality, several strategic opportunities emerge for strengthening Netflix's content investments, user experience, and evaluation systems.

First, CPS results clearly indicate that Adventure, Comedy, History, and U.S.-produced titles consistently generate the strongest performance, while TV Series outperform Movies due to longer watch durations and stronger retention. Netflix should therefore prioritize investment in these high-value genres and expand long-form episodic content, especially multi-season franchises. Because completion rates cluster around 55–60% regardless of genre, improvement should focus on execution—optimizing episode pacing, metadata, thumbnails, and early discovery surfaces. Titles that simultaneously achieve high CPS and high completion should be promoted and considered for sequels or expanded universes.

Second, user behavior shows that large-screen devices such as Smart TVs and Desktops drive the majority of watch minutes, while Tablets significantly underperform. Enhancing the large-screen viewing experience—through UI refinement, personalization, and quality-of-streaming upgrades—will amplify engagement among core users. High-UES viewers, who are typically Standard or Premium subscribers and heavy consumers of TV Series, represent Netflix's most valuable audience segment. To deepen loyalty, Netflix should tailor recommendations and marketing to their preferred genres, strengthen multi-device continuity, and introduce premium-tier features or early-access content targeting this high-engagement group. Search is another powerful engagement engine, with a 90% conversion rate. Improving search by adding smart filter suggestions, refining query understanding, and addressing low-conversion keywords will further accelerate content discovery.

Finally, the review system requires modernization. With 99.7% of 5-star ratings coming from unverified views, raw star ratings significantly overestimate satisfaction. Netflix should shift toward behavior-based metrics—CPS, UES, completion rate, and verified-view signals—when evaluating content performance or ranking titles in recommendations. Introducing watch-progress thresholds before allowing users to rate, and incorporating more granular rating options for episodic content, would greatly improve sentiment reliability. Aligning personalization algorithms with CPS × UES patterns will ensure that strong content reaches high-value viewers and that rising engagement trends are used as early indicators for retention, upsell, and marketing decisions.

Taken together, these recommendations will help Netflix optimize content ROI, improve user satisfaction, and deepen engagement by aligning production, discovery, and evaluation around the platform's most valuable genres, formats, and users.

9. Appendix — SQL Code

```
----------Yearly Movie vs TV Show Trend----------

SELECT
        release_year,
        content_type,
        SUM(title_count) AS total_titles
FROM    SM_NM_YEARLY_CONTENT_STATS_VW
WHERE   content_type in ('Movie','TV Series')
GROUP   BY
        release_year,
        content_type
ORDER   BY
        release_year,
        content_type;

----------Country and Genre Contribution----------

-- Count titles by country and primary genre
SELECT
        c.country_name,
        g.genre_name            AS primary_genre,
        COUNT(*)                AS title_count
FROM    SM_NM_MOVIE         m
        JOIN SM_NM_COUNTRY c ON m.country_id        = c.country_id
        JOIN SM_NM_GENRE   g ON m.genre_primary_fk = g.genre_id
GROUP   BY
        c.country_name,
        g.genre_name
ORDER   BY
        c.country_name,
        title_count DESC;

----------Duration and Rating Relationship----------

SELECT
        duration_minute,
        rating
FROM    SM_NM_MOVIE m
        JOIN SM_NM_REVIEW r ON m.movie_id = r.movie_id
WHERE   duration_minute IS NOT NULL
```

```
   AND    rating IS NOT NULL;

----------Content Performance Score (CPS)----------

-- Step 1: Review statistics per movie
WITH review_stats AS (
      SELECT
            r.movie_id,
            AVG(r.rating) AS avg_rating,
            COUNT(*)          AS total_reviews,
            SUM(
                  CASE
                        WHEN EXISTS (
                              SELECT 1
                              FROM      SM_NM_WATCH_HISTORY w
                              WHERE    w.movie_id = r.movie_id
                              AND       w.user_id    = r.user_id
                        )
                        THEN 1 ELSE 0
                  END
            ) AS verified_reviews
      FROM     SM_NM_REVIEW r
      GROUP   BY
            r.movie_id
),

-- Step 2: Watch statistics per movie
watch_stats AS (
      SELECT
            movie_id,
            SUM(watch_duration_minute) AS total_watch_minute
      FROM     SM_NM_WATCH_HISTORY
      GROUP   BY
            movie_id
),

-- Step 3: Combine data and compute verified_review_ratio
movie_metrics AS (
      SELECT
            m.movie_id,
            NVL(rs.avg_rating, 0)                  AS avg_rating,
            NVL(ws.total_watch_minute, 0)    AS total_watch_minute,
            CASE
                  WHEN NVL(rs.total_reviews, 0) = 0 THEN 0
```

```
                    ELSE rs.verified_reviews / rs.total_reviews
                END AS verified_review_ratio
        FROM    SM_NM_MOVIE m
                LEFT JOIN review_stats rs ON m.movie_id = rs.movie_id
                LEFT JOIN watch_stats  ws ON m.movie_id = ws.movie_id
),

-- Step 4: Normalize metrics to [0,1]
movie_metrics_norm AS (
        SELECT
                movie_id,
                avg_rating,
                total_watch_minute,
                verified_review_ratio,
                avg_rating
                  / NULLIF(MAX(avg_rating) OVER (), 0)                  AS norm_rating,
                total_watch_minute
                  / NULLIF(MAX(total_watch_minute) OVER (), 0)          AS norm_watch_time,
                verified_review_ratio
                  / NULLIF(MAX(verified_review_ratio) OVER (), 0)       AS norm_verified_ratio
        FROM    movie_metrics
)

-- Step 5: Compute CPS score using final weights
SELECT
        movie_id,
        0.60 * norm_rating
      + 0.25 * norm_watch_time
      + 0.15 * norm_verified_ratio  AS content_performance_score
FROM    movie_metrics_norm
ORDER   BY
        content_performance_score DESC;

----------Base Content Insights Dataset----------
-- Per-title CPS with key business attributes:
-- title, content type, primary genre, country, original flag, year, duration

SELECT
        m.movie_id,
        m.title,
        ct.content_type,
        g.genre_name                AS primary_genre,
        c.country_name,
        m.is_netflix_original,
```

```
        m.release_year,
        m.duration_minute,
        cps.content_performance_score AS cps
FROM    SM_NM_MOVIE_CPS_MV          cps
        JOIN SM_NM_MOVIE            m    ON cps.movie_id = m.movie_id
        JOIN SM_NM_CONTENT_TYPE     ct ON m.content_id = ct.content_id
        JOIN SM_NM_GENRE            g    ON m.genre_primary_fk = g.genre_id
        JOIN SM_NM_COUNTRY          c    ON m.country_id = c.country_id
ORDER   BY
        cps DESC;

----------Overall Search Volume and Intensity----------

-- Total number of searches and average searches per user

SELECT
        COUNT(*)                                AS total_searches,
        COUNT(DISTINCT user_id)                 AS distinct_users,
        ROUND(COUNT(*) / NULLIF(COUNT(DISTINCT user_id), 0), 2)
          AS avg_searches_per_user
FROM    SM_NM_SEARCH_LOG;

----------Zero-result Search Rate----------

SELECT
        COUNT(*)                                                AS total_searches,
        SUM(CASE WHEN results_returned = 0 THEN 1 ELSE 0 END)
          AS zero_result_searches,
        ROUND(
          SUM(CASE WHEN results_returned = 0 THEN 1 ELSE 0 END)
          / NULLIF(COUNT(*), 0),
          4
        ) * 100                                                 AS zero_result_pct
FROM    SM_NM_SEARCH_LOG;

----------Top Used Filters----------

SELECT
        f.used_filter,
        COUNT(*) AS times_used
FROM    SM_NM_SEARCH_LOG    s
        JOIN SM_NM_USED_FILTER f
          ON s.used_filter_id = f.used_filter_id
GROUP   BY
```

```
        f.used_filter
ORDER   BY
        times_used DESC;

----------Frequently Searched Queries with Zero Results----------

SELECT *
FROM (
        SELECT
                LOWER(TRIM(search_query)) AS normalized_query,
                COUNT(*)                  AS search_count
        FROM    SM_NM_SEARCH_LOG
        WHERE   results_returned = 0
        GROUP   BY
                LOWER(TRIM(search_query))
        HAVING COUNT(*) >= 5
        ORDER   BY
                search_count DESC
      )
WHERE ROWNUM <= 20;

----------Search Duration in January 2025----------

WITH daily_searches AS (
      SELECT
            search_date,
            COUNT(*) AS daily_search_count
      FROM SM_NM_SEARCH_LOG
      WHERE search_date >= DATE '2025-01-01'
        AND search_date <   DATE '2025-02-01'
      GROUP BY search_date
)
SELECT
      search_date,
      daily_search_count
FROM daily_searches
ORDER BY search_date;

----------Gender Distribution Across Subscription Plans----------

SELECT
        sp.subscription_plan,
        g.gender_name,
        COUNT(*) AS user_count,
```

```
        ROUND(
            COUNT(*) * 100
            / SUM(COUNT(*)) OVER (PARTITION BY sp.subscription_plan),
            1
        ) AS pct_within_plan
FROM    SM_NM_USER u
        JOIN SM_NM_GENDER g ON u.gender_id = g.gender_id
        JOIN SM_NM_SUBSCRIPTION_PLAN sp ON u.subscription_plan_id =
sp.subscription_plan_id
GROUP   BY
        sp.subscription_plan,
        g.gender_name
ORDER   BY
        sp.subscription_plan,
        user_count DESC;

----------Overall Gender Distribution----------

SELECT
        g.gender_name,
        COUNT(*) AS user_count,
        ROUND(
            COUNT(*) * 100
            / SUM(COUNT(*)) OVER (),
            1
        ) AS pct_overall
FROM    SM_NM_USER u
        JOIN SM_NM_GENDER g
            ON u.gender_id = g.gender_id
GROUP   BY
        g.gender_name
ORDER   BY
        user_count DESC;

----------Top 3 Device Types by Viewing Duration----------

-- Step 1: Aggregate total watch duration per device type
WITH device_watch AS (
    SELECT
            dt.device_type,
            SUM(wh.watch_duration_minute) AS total_watch_minutes
    FROM    SM_NM_WATCH_HISTORY wh
            JOIN SM_NM_DEVICE_TYPE dt
                ON wh.device_type_id = dt.device_type_id
```

```
        GROUP BY
                dt.device_type
)

-- Step 2: Return Top 3 device types with highest total viewing duration
SELECT
        device_type,
        total_watch_minutes
FROM    device_watch
ORDER BY
        total_watch_minutes DESC
FETCH FIRST 3 ROWS ONLY;

----------User Engagement Score (UES) Updated----------

-- Step 1:
WITH user_base AS (
        SELECT user_id
        FROM    SM_NM_USER
),

-- Step 2:
user_watch AS (
        SELECT
                wh.user_id,
                SUM(wh.watch_duration_minute)       AS total_watch_minutes,
                COUNT(DISTINCT wh.device_type_id) AS device_diversity
        FROM    SM_NM_WATCH_HISTORY wh
        GROUP   BY wh.user_id
),

-- Step 3:
user_search AS (
        SELECT
                sl.user_id,
                COUNT(*) AS search_count
        FROM    SM_NM_SEARCH_LOG sl
        GROUP   BY sl.user_id
),

-- Step 4:
user_metrics AS (
        SELECT
                ub.user_id,
```

```
        NVL(uw.total_watch_minutes, 0) AS total_watch_minutes,
        NVL(uw.device_diversity,    0) AS device_diversity,
        NVL(us.search_count,        0) AS search_count
    FROM    user_base ub
        LEFT JOIN user_watch    uw ON ub.user_id = uw.user_id
        LEFT JOIN user_search us ON ub.user_id = us.user_id
),

-- Step 5:
user_metrics_norm AS (
    SELECT
        user_id,
        total_watch_minutes
            / NULLIF(MAX(total_watch_minutes) OVER (), 0) AS watch_minutes_norm,
        device_diversity
            / NULLIF(MAX(device_diversity) OVER (), 0)        AS device_diversity_norm,
        search_count
            / NULLIF(MAX(search_count) OVER (), 0)            AS search_activity_norm
    FROM    user_metrics
),

-- Step 6:
user_ues AS (
    SELECT
        user_id,
        0.65 * watch_minutes_norm     +
        0.25 * device_diversity_norm +
        0.10 * search_activity_norm     AS user_engagement_score
    FROM    user_metrics_norm
)

-- Step 7:
SELECT
        user_id,
        user_engagement_score
FROM    user_ues
ORDER   BY user_engagement_score DESC;

----------Top 20 UES Users – Full Behavior & Preference Profile----------

WITH top_ues_users AS (
    SELECT
        user_id,
        user_engagement_score
```

```
    FROM      SM_NM_USER_UES_MV
    ORDER    BY user_engagement_score DESC
    FETCH FIRST 20 ROWS ONLY
),

user_profile AS (
    SELECT
            u.user_id,
            sp.subscription_plan
    FROM      SM_NM_USER u
            JOIN SM_NM_SUBSCRIPTION_PLAN sp
               ON u.subscription_plan_id = sp.subscription_plan_id
),

user_device AS (
    SELECT
            user_id,
            COUNT(DISTINCT device_type_id) AS device_diversity,
            MAX(device_type) KEEP (DENSE_RANK LAST ORDER BY total_watch_minutes) AS
primary_device
    FROM (
        SELECT
                wh.user_id,
                wh.device_type_id,
                dt.device_type,
                SUM(wh.watch_duration_minute) AS total_watch_minutes
        FROM      SM_NM_WATCH_HISTORY wh
                JOIN SM_NM_DEVICE_TYPE dt
                   ON wh.device_type_id = dt.device_type_id
        GROUP    BY
                wh.user_id, wh.device_type_id, dt.device_type
    )
    GROUP BY user_id
),

user_content AS (
    SELECT
            wh.user_id,
            ct.content_type,
            g.genre_name,
            c.country_name,
            COUNT(*) AS view_count
    FROM      SM_NM_WATCH_HISTORY wh
            JOIN SM_NM_MOVIE m
```

```
            ON wh.movie_id = m.movie_id
        JOIN SM_NM_CONTENT_TYPE ct
            ON m.content_id = ct.content_id
        JOIN SM_NM_GENRE g
            ON m.genre_primary_fk = g.genre_id
        JOIN SM_NM_COUNTRY c
            ON m.country_id = c.country_id
    GROUP BY
        wh.user_id, ct.content_type, g.genre_name, c.country_name
),

user_content_main AS (
    SELECT
        user_id,
        MAX(content_type) KEEP (DENSE_RANK LAST ORDER BY view_count) AS primary_content_type,
        MAX(genre_name) KEEP (DENSE_RANK LAST ORDER BY view_count) AS primary_genre,
        MAX(country_name) KEEP (DENSE_RANK LAST ORDER BY view_count) AS primary_country
    FROM user_content
    GROUP BY user_id
),

user_search AS (
    SELECT
        sl.user_id,
        COUNT(*) AS search_count
    FROM SM_NM_SEARCH_LOG sl
    GROUP BY sl.user_id
),

user_watch AS (
    SELECT
        wh.user_id,
        SUM(wh.watch_duration_minute) AS total_watch_minutes,
        COUNT(*) AS watch_sessions,
        AVG(wh.progress_percentage) AS avg_completion_pct
    FROM SM_NM_WATCH_HISTORY wh
    GROUP BY wh.user_id
)

SELECT
    t.user_id,
```

```
        t.user_engagement_score          AS ues,
        up.subscription_plan,
        uw.total_watch_minutes,
        uw.watch_sessions,
        ROUND(uw.avg_completion_pct, 1) AS avg_completion_pct,
        ud.device_diversity,
        ud.primary_device,
        NVL(us.search_count, 0) AS search_count,
        ucm.primary_content_type,
        ucm.primary_genre,
        ucm.primary_country
FROM    top_ues_users t
        LEFT JOIN user_profile          up   ON t.user_id = up.user_id
        LEFT JOIN user_watch             uw   ON t.user_id = uw.user_id
        LEFT JOIN user_device            ud   ON t.user_id = ud.user_id
        LEFT JOIN user_search            us   ON t.user_id = us.user_id
        LEFT JOIN user_content_main ucm ON t.user_id = ucm.user_id
ORDER   BY t.user_engagement_score DESC;

----------CPS × UES – Viewers of Top 20 CPS Movies----------

WITH top_cps_movies AS (
    SELECT
        movie_id,
        content_performance_score AS cps
    FROM    SM_NM_MOVIE_CPS_MV
    ORDER   BY content_performance_score DESC
    FETCH FIRST 20 ROWS ONLY
),
movie_watch AS (
    SELECT
        wh.movie_id,
        wh.user_id
    FROM    SM_NM_WATCH_HISTORY wh
        JOIN top_cps_movies tm
          ON wh.movie_id = tm.movie_id
    GROUP   BY wh.movie_id, wh.user_id
),
movie_ues AS (
    SELECT
        mw.movie_id,
        AVG(ues.user_engagement_score) AS avg_viewer_ues,
        COUNT(DISTINCT mw.user_id)        AS unique_viewers
    FROM    movie_watch mw
```

```
                JOIN SM_NM_USER_UES_MV ues
                  ON mw.user_id = ues.user_id
    GROUP   BY mw.movie_id
)
SELECT
        m.movie_id,
        m.title,
        tm.cps,
        mu.unique_viewers,
        ROUND(mu.avg_viewer_ues, 3) AS avg_viewer_ues
FROM    top_cps_movies tm
        JOIN SM_NM_MOVIE m      ON tm.movie_id = m.movie_id
        LEFT JOIN movie_ues mu ON tm.movie_id = mu.movie_id
ORDER   BY tm.cps DESC;

----------CPS × UES – Content Watched by Top 20 UES Users----------

WITH top_ues_users AS (
    SELECT
            user_id,
            user_engagement_score AS ues
    FROM    SM_NM_USER_UES_MV
    ORDER   BY user_engagement_score DESC
    FETCH FIRST 20 ROWS ONLY
),
user_movies AS (
    SELECT DISTINCT
            wh.user_id,
            wh.movie_id
    FROM    SM_NM_WATCH_HISTORY wh
            JOIN top_ues_users tu
              ON wh.user_id = tu.user_id
),
movie_cps AS (
    SELECT
            mc.movie_id,
            mc.content_performance_score AS cps
    FROM    SM_NM_MOVIE_CPS_MV mc
)
SELECT
        tu.user_id,
        tu.ues,
        AVG(mc.cps)                         AS avg_cps_watched,
        MIN(mc.cps)                         AS min_cps_watched,
```

```
        MAX(mc.cps)                                 AS max_cps_watched,
        COUNT(DISTINCT um.movie_id)         AS distinct_titles_watched
FROM    top_ues_users tu
        JOIN user_movies um ON tu.user_id = um.user_id
        JOIN movie_cps    mc ON um.movie_id = mc.movie_id
GROUP   BY tu.user_id, tu.ues
ORDER   BY tu.ues DESC;

----------Rating Distribution – 5-star vs 1-star----------

WITH rating_stats AS (
    SELECT
            r.rating,
            COUNT(*) AS review_count
    FROM    SM_NM_REVIEW r
    WHERE   r.rating IN (1, 5)
    GROUP   BY r.rating
),
total_extreme AS (
    SELECT SUM(review_count) AS total_reviews
    FROM    rating_stats
)
SELECT
        CASE
            WHEN rs.rating = 5 THEN '5-star'
            WHEN rs.rating = 1 THEN '1-star'
        END AS rating_band,
        rs.review_count,
        ROUND(rs.review_count / te.total_reviews * 100, 1) AS pct_of_extreme_reviews
FROM    rating_stats rs
        CROSS JOIN total_extreme te
ORDER   BY rs.rating DESC;

----------Share of Verified View 5-Star Reviews----------

WITH five_star_reviews AS (
    SELECT
            r.review_id,
            r.user_id,
            r.movie_id,
            r.rating
    FROM    SM_NM_REVIEW r
    WHERE   r.rating = 5
),
```

```
five_star_with_flag AS (
    SELECT
        fs.review_id,
        fs.user_id,
        fs.movie_id,
        CASE
            WHEN EXISTS (
                SELECT 1
                FROM   SM_NM_WATCH_HISTORY wh
                WHERE  wh.user_id  = fs.user_id
                AND    wh.movie_id = fs.movie_id
                AND    wh.progress_percentage >= 70
            ) THEN 'VERIFIED_VIEW'
            ELSE 'UNVERIFIED_VIEW'
        END AS verified_view_flag
    FROM   five_star_reviews fs
)

SELECT
    verified_view_flag,
    COUNT(*) AS review_count,
    ROUND(
        COUNT(*) / SUM(COUNT(*)) OVER () * 100,
        1
    ) AS pct_of_5star_reviews
FROM   five_star_with_flag
GROUP  BY verified_view_flag
ORDER  BY verified_view_flag;

----------Recommendation Consumption by Type----------

SELECT
    r.recommendation_type_id,
    rt.recommendation_type,
    COUNT(*) AS total_recommended,
    SUM(
        CASE
            WHEN wh.session_id IS NOT NULL THEN 1
            ELSE 0
        END
    ) AS watched_after_recommendation,
    ROUND(
        SUM(
```

```
                    CASE
                        WHEN wh.session_id IS NOT NULL THEN 1
                        ELSE 0
                    END
                ) / COUNT(*) * 100,
                2
            ) AS consumption_rate_pct
FROM      SM_NM_RECOMMENDATION              r
            LEFT JOIN SM_NM_RECOMMENDATION_TYPE rt
                    ON r.recommendation_type_id = rt.recommendation_type_id
            LEFT JOIN SM_NM_WATCH_HISTORY             wh
                    ON r.user_id    = wh.user_id
                   AND r.movie_id = wh.movie_id
GROUP    BY
            r.recommendation_type_id,
            rt.recommendation_type
ORDER    BY
            r.recommendation_type_id;

----------------------CPS + Completion Metrics Final Model----------------------

/* Step 1: Review statistics per movie */
WITH review_stats AS (
      SELECT
                  r.movie_id,
                  AVG(r.rating) AS avg_rating,
                  COUNT(*)           AS total_reviews,
                  SUM(
                        CASE
                              WHEN EXISTS (
                                    SELECT 1
                                    FROM      SM_NM_WATCH_HISTORY w
                                    WHERE    w.movie_id = r.movie_id
                                    AND       w.user_id    = r.user_id
                              ) THEN 1
                              ELSE 0
                        END
                  ) AS verified_reviews
      FROM      SM_NM_REVIEW r
      GROUP    BY r.movie_id
),

/* Step 2: Watch statistics per movie (total watch minutes for CPS) */
watch_stats AS (
```

```
    SELECT
            movie_id,
            SUM(watch_duration_minute) AS total_watch_minute
    FROM    SM_NM_WATCH_HISTORY
    GROUP   BY movie_id
),

/* Step 3: Combine metrics + compute verified_review_ratio */
movie_metrics AS (
    SELECT
            m.movie_id,
            NVL(rs.avg_rating, 0)               AS avg_rating,
            NVL(ws.total_watch_minute, 0) AS total_watch_minute,
            CASE
                WHEN NVL(rs.total_reviews, 0) = 0 THEN 0
                ELSE rs.verified_reviews / rs.total_reviews
            END AS verified_review_ratio
    FROM    SM_NM_MOVIE m
            LEFT JOIN review_stats rs ON m.movie_id = rs.movie_id
            LEFT JOIN watch_stats    ws ON m.movie_id = ws.movie_id
),

/* Step 4: Normalize metrics to [0,1] */
movie_metrics_norm AS (
    SELECT
            movie_id,
            avg_rating,
            total_watch_minute,
            verified_review_ratio,
            avg_rating
                / NULLIF(MAX(avg_rating) OVER (), 0)                    AS norm_rating,
            total_watch_minute
                / NULLIF(MAX(total_watch_minute) OVER (), 0)        AS norm_watch_time,
            verified_review_ratio
                / NULLIF(MAX(verified_review_ratio) OVER (), 0)     AS norm_verified_ratio
    FROM    movie_metrics
),

/* Step 5: Compute CPS using final weights (0.60 / 0.25 / 0.15) */
movie_cps AS (
    SELECT
            movie_id,
            0.60 * norm_rating
          + 0.25 * norm_watch_time
```

```
            + 0.15 * norm_verified_ratio AS cps
    FROM    movie_metrics_norm
),

/* Step 6: Completion statistics per movie */
completion_stats AS (
    SELECT
            wh.movie_id,
            AVG(wh.progress_percentage) AS avg_completion_pct,
            COUNT(DISTINCT wh.user_id) AS total_viewers
    FROM    SM_NM_WATCH_HISTORY wh
    GROUP   BY wh.movie_id
)

/* Step 7: Final combined dataset */
SELECT
        m.movie_id,
        m.title,
        ct.content_type,
        g.genre_name AS primary_genre,
        c.country_name,
        m.is_netflix_original,
        m.release_year,
        m.duration_minute,
        cs.avg_completion_pct,
        cs.total_viewers,
        mc.cps AS content_performance_score
FROM    SM_NM_MOVIE             m,
        SM_NM_CONTENT_TYPE      ct,
        SM_NM_GENRE             g,
        SM_NM_COUNTRY           c,
        completion_stats        cs,
        movie_cps               mc
WHERE   m.movie_id          = cs.movie_id
AND     m.movie_id          = mc.movie_id
AND     m.content_id        = ct.content_id
AND     m.genre_primary_fk  = g.genre_id
AND     m.country_id        = c.country_id
ORDER   BY
        cs.avg_completion_pct DESC
FETCH FIRST 50 ROWS ONLY
```